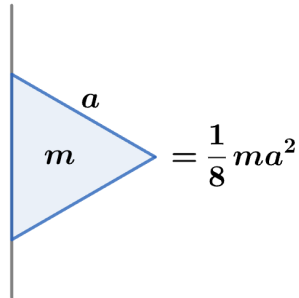


2016 F=ma Exam: Problem 24

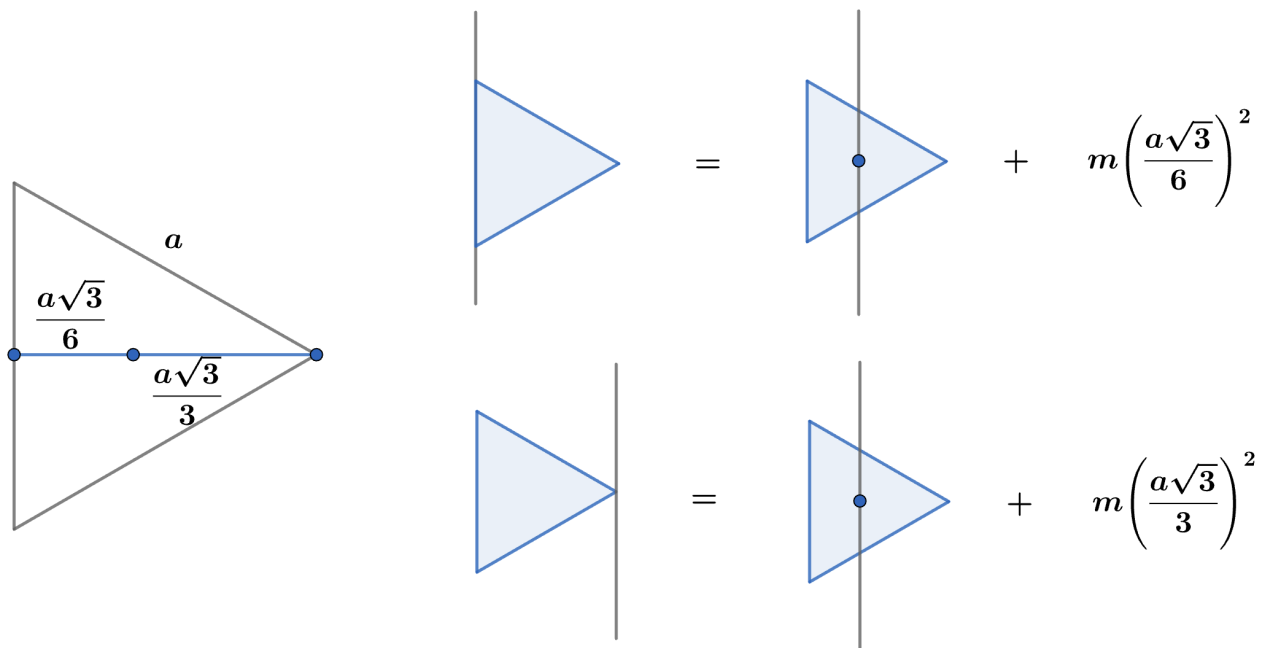
Kevin S. Huang



We are given that the moment of inertia I_1 of an equilateral triangle (axis through side) is

$$I_1 = \frac{1}{8}ma^2$$

Let's first find the moment of inertia I_2 of an equilateral triangle (axis through vertex). We use the parallel axis theorem:



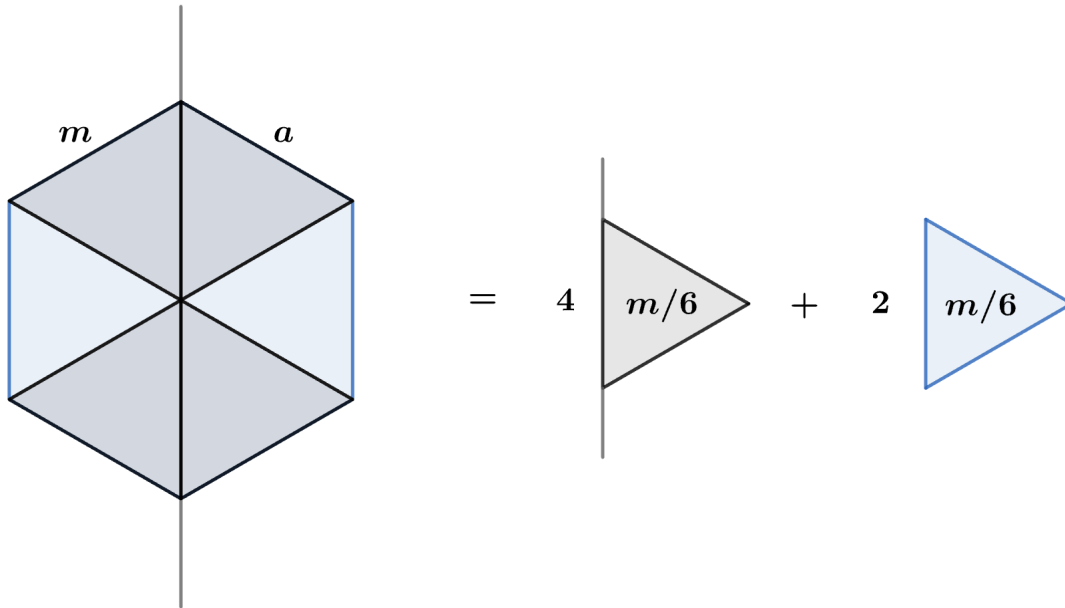
$$I_1 = I_{\text{CM}} + m \left(\frac{a\sqrt{3}}{6} \right)^2 = I_{\text{CM}} + \frac{ma^2}{12}$$

$$I_2 = I_{\text{CM}} + m \left(\frac{a\sqrt{3}}{3} \right)^2 = I_{\text{CM}} + \frac{ma^2}{3}$$

Subtracting these two equations,

$$I_2 - I_1 = ma^2 \left(\frac{1}{3} - \frac{1}{12} \right) = \frac{ma^2}{4}$$

$$I_2 = I_1 + \frac{1}{4}ma^2 = \left(\frac{1}{8} + \frac{1}{4} \right) ma^2 = \frac{3}{8}ma^2$$



The hexagon can be broken down into 6 smaller equilateral triangles. Thus, the moment of inertia about an axis through opposite vertices of a hexagon is equivalent to the moment of inertia of 4 triangles (axis through side) and 2 triangles (axis through vertex). If m is the mass of the hexagon, then each triangle has mass $m/6$ so

$$I_{\text{hex}} = 4 \left(\frac{I_1}{6} \right) + 2 \left(\frac{I_2}{6} \right) = \frac{2}{3} \left(\frac{1}{8}ma^2 \right) + \frac{1}{3} \left(\frac{3}{8}ma^2 \right) = \frac{5}{24}ma^2$$

Hence, the answer is B.